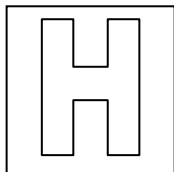


Name: SUGGESTED ANSWERS	Index No.:	CT Group: 11
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PIONEER JUNIOR COLLEGE

2012 JC2 PRELIMINARY EXAMINATION
HIGHER 1



CHEMISTRY

8872/02

10 September 2012

Candidates answer Section A on the Question Paper

Additional Materials: Data Booklet
Writing Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, index no and CT group on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **two** questions on separate answer paper.

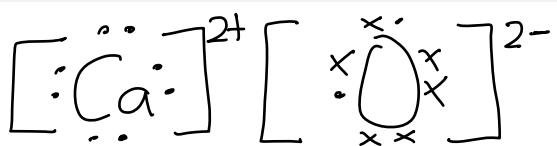
At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE			
Section A		Section B	
1	/ 13	4	/ 20
2	/ 16	5	/ 20
3	/ 11	6	/ 20
		Penalty	s.f. / units
		TOTAL	/ 80

Section A (40 marks)

Answer all questions in the spaces provided.

1	(a)	Iron is the most common element (by mass) forming the planet Earth as a whole, forming much of Earth's outer and inner core. Iron is the most widely used of all the metals, accounting for 95% of worldwide metal production.
	(i)	By means of the structure and bonding, explain why iron has a higher melting point than calcium.
		Both iron and calcium consist of giant metallic structure with metallic bonding, which comprises of electrostatic forces of attraction found between cations and mobile/delocalized electrons. There is a higher charge density in Fe^{2+} compared to Ca^{2+} , hence resulting in stronger electrostatic forces of attraction in iron than calcium, thus more energy is required to overcome them, therefore leading to a high melting point in iron.
	(ii)	Draw dot-and-cross diagram to show the electron arrangement in calcium oxide.
		
		[3]
	(b)	Iron has several uses in industries such as being used as catalyst in the Haber Process to synthesise ammonia. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) \quad \Delta H = -92 \text{ kJ mol}^{-1}$ The optimal conditions for the synthesis of ammonia in industries are listed below. Temperature : 450 – 500 °C Pressure : 200 – 250 atm
	(i)	Using <i>Le Chatelier's principle</i> , explain why a higher temperature and pressure is not used instead.
		By Le Chatelier's Principle, by using higher temperature, position of equilibrium of an exothermic reaction shifts to the <u>left, to decrease the temperature</u>, giving a <u>low yield</u> of ammonia. However, the <u>rate of reaction</u> would be too slow at low temperature hence a reasonable temperature of 450-500°C is employed. By Le Chatelier's Principle, increasing pressure will shift equilibrium position to the <u>side where lesser number of moles of gases are produced</u>. Hence position of eqm will shift to <u>the right</u>, favouring the production of NH_3. However, with increasing pressure, the <u>production cost increases</u> (e.g. pipes and reaction vessel have to be made thicker to withstand the pressure).

- (ii) A 1 dm³ reaction vessel contained 1.36 moles of N₂, 1.84 moles of H₂ and 0.142 moles of NH₃ at equilibrium at 1000K.

Calculate the K_c , stating its units, and hence calculate the amount of N₂ that is required to be added into the system such that the final number of moles of NH₃ will be increased to 0.150 moles at 1000K.

$$\begin{aligned} K_c &= [\text{NH}_3]^2 / [\text{N}_2][\text{H}_2]^3 \\ &= (0.142)^2 / (1.36)(1.84)^3 \\ &= 0.00238 \text{ mol}^{-2} \text{ dm}^6 \end{aligned}$$

Let x be the no. of moles of nitrogen to be added.

	N ₂ (g)	+	3H ₂ (g)	⇌	2NH ₃ (g)
Initial amount/ mol	1.36+x		1.84		0.142
Change in amount/ mol	-0.0040		-0.0120		+0.0080
Final amount at eqm/ mol	1.356+x		1.828		0.150

$$\begin{aligned} K_c &= (0.150^2) / (1.828^3)(1.356+x) \\ x &= 0.1917 \\ &= 0.192 \end{aligned}$$

[6]

- (c) Iron plays an important role in biology, forming complexes with molecular oxygen in haemoglobin in the blood to transport oxygen in the blood.

Another important component of human blood consists of the weak acid H₂CO₃ and conjugate base HCO₃⁻, which aids in maintaining the pH of blood.

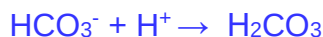
- (i) Define what is meant by pH of a solution.

$$\text{pH} = -\lg [\text{H}^+] \text{ or}$$

pH is the negative logarithm to base 10 of the hydrogen ion concentration in mol dm⁻³.

- (ii) 1 mole of H₂CO₃ and 1 mole of HCO₃⁻ were dissolved in 1 dm³ water to produce a buffer solution. 1x10⁻² moles of HCl was then added to the system. Calculate the new concentration of H₂CO₃ and HCO₃⁻ in the buffer.

When H⁺ is added:



$$\begin{aligned} \text{New concentration of HCO}_3^- &= 1 - 0.01 \\ &= 0.990 \text{ mol dm}^{-3} \end{aligned}$$

$$\text{New concentration of H}_2\text{CO}_3 = 1.01 \text{ mol dm}^{-3}$$

(iii) Given that



Calculate the new concentration of H^+ in the buffer above, and hence calculate the final pH of the buffer solution.

$$K_a = \frac{[\text{HCO}_3^-][\text{H}^+]}{[\text{H}_2\text{CO}_3]}$$

$$4.5 \times 10^{-7} = (0.99)[\text{H}^+] / (1.01)$$

$$[\text{H}^+] = 4.59 \times 10^{-7}$$

$$\text{pH} = -\lg [\text{H}^+]$$

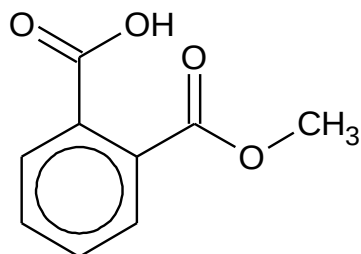
$$= 6.34$$

[4]

[Total: 13]

- 2 Aspirin, also known as acetylsalicylic acid, is a salicylate drug, often used as an analgesic to relieve minor aches and pains, and as an anti-inflammatory medication.

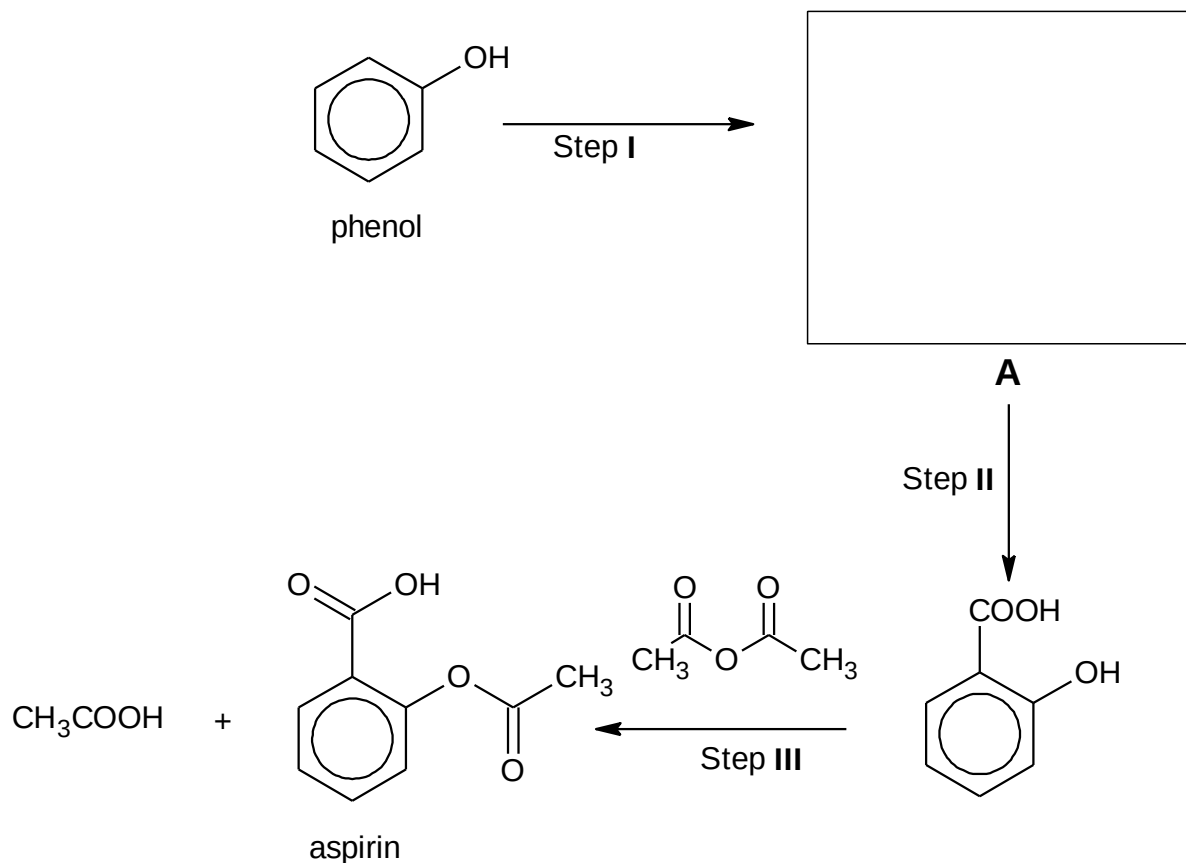
Acetylsalicylic acid has the following structural formula:



- (a) Other than the phenyl ring, list down two other functional groups found in aspirin.
Ester, carboxylic acid

[1]

(b) Aspirin can be synthesized from phenol by the following reaction scheme.

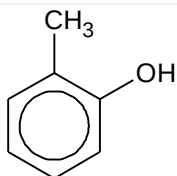


(i) Suggest reagents and conditions for Steps I and II. You may assume in your working that the hydroxyl group does not take part in the reaction.

Step I: anhydrous $AlCl_3$, CH_3Cl , heat

Step II: $KMnO_4$, $H_2SO_4(aq)$, heat

(ii) Draw the structure for A in the box above.



(ii) State the type of reaction taking place in Steps I and III.

Step I : substitution

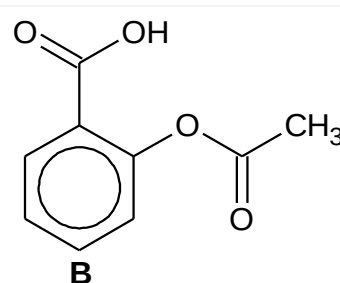
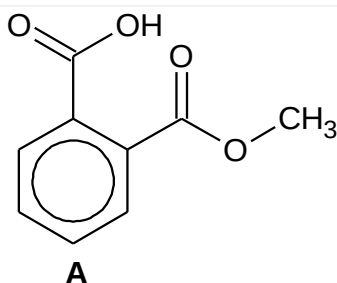
Step III: condensation /substitution

- (c) Aspirin has a pK_a value of 3.48, while benzoic acid has a pK_a value of 4.20. Explain why this is so.

The ester functional group in aspirin is electron withdrawing, hence helps to disperse the negative charge on the carboxylate anion of aspirin, thus anion of aspirin is more stable than benzoic acid anion. The **more stable** the anion, the **stronger** the acid.

[2]

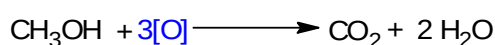
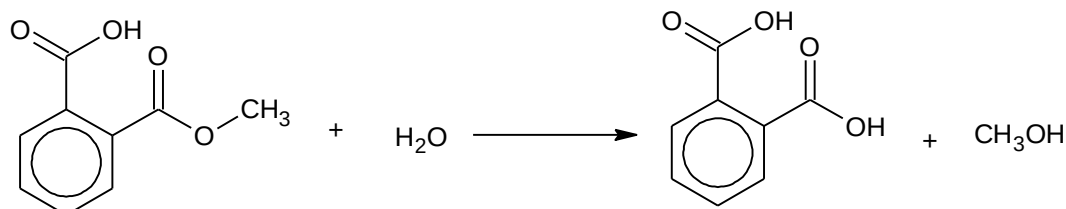
- (d) (i) Describe a chemical test to distinguish between the two compounds. State clearly how each compound behaves in the test. Write equations for any relevant reactions.



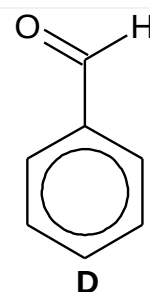
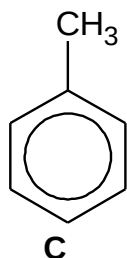
$KMnO_4$, H_2SO_4 , heat

Purple $KMnO_4$ decolourised for compound **A**. Colourless gas evolved, CO_2 , which forms white ppt in lime water.

No observable change for compound **B**.



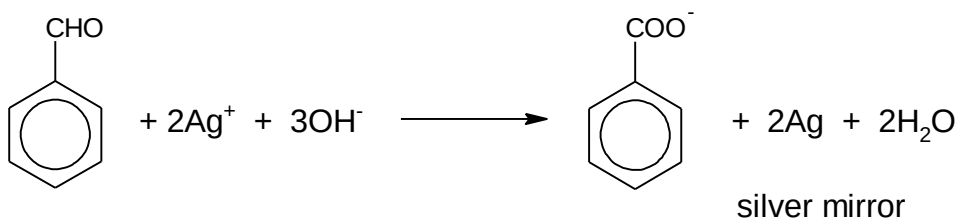
- (ii)



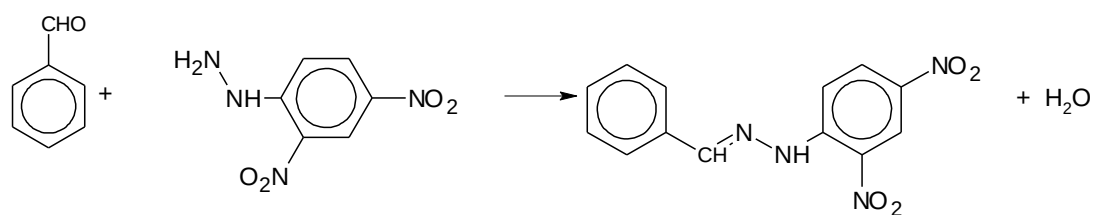
Tollens' reagent, warm

Silver mirror observed for compound **D**,

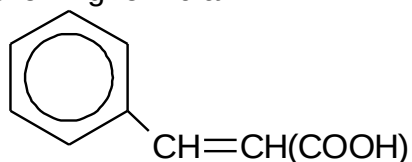
No observable change for **C**.



2,4-DNPH

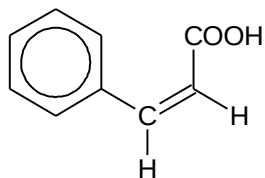
Orange ppt observed for **D**.No observable change for **C**.

[6]

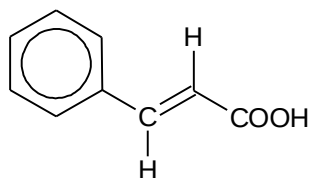
(e) A carboxylic acid has the following formula.

State the type of isomerism that it can display and draw the structural formulae of its isomers.

Geometric isomerism



cis isomer



trans isomer

[2]

[Total:16]

- 3 (a)** In an automobile engine, petrol (gasoline) vapour and air are ignited by an electric spark, producing an explosive reaction which drives the pistons of the engine. Petrol vapour is chiefly a mixture of C_5 to C_{10} alkanes. It is important that the combustions take place smoothly for efficient running of a high performance engine. Apart from the usage of octane as fuel, ethanol is now used as a low-cost alternative fuel. Ethanol is considered to be a better fuel for the environment than octane as ethanol-fueled vehicles produce lower carbon monoxide and carbon dioxide emissions.

The table below shows the sources and the enthalpy change of combustion of ethanol and octane used in vehicles.

Fuel	Main Sources	Enthalpy change of combustion / kJ mol^{-1}
Octane	Refined from crude oil	- 5461
Ethanol	Hydration of ethene Fermentation	- 1366

Several studies were conducted to analyse the sources of air pollution. These studies ultimately attributed a significant portion of air pollution to the automobile. Hence analysis was done and the typical composition of the gases in petrol engine exhaust fumes is as follows:

Petrol	
Compound	Percentage %
CO_2	14
H_2O	12
CO	1-2
NO_x	0.3
C_xH_y	0.2
SO_2	Possible traces

It was found that the emission of gases from the automobile has a variety of negative effects on public health and the natural environment. This forced the invention of catalytic converter.

A catalytic converter is a vehicle emissions control device which converts toxic byproducts of combustion in the exhaust of an internal combustion engine to less toxic substances by catalysed chemical reactions. The specific reactions vary with the type of catalyst installed. Most present-day vehicles that run on gasoline are fitted with a "three way" converter, so named because it converts the three main pollutants in automobile exhaust into harmless products.

- (a)** A typical motor car requires 6 dm^3 of octane to travel 100 km.

- (i)** Construct an equation for the complete combustion of octane.



- (ii)** The density of octane is 0.70 g cm^{-3} .
Calculate the number of moles of octane present in the 6 dm^3 of octane.

		No of moles of octane present = $(0.7 \times 6 \times 1000) / 114$ $= 36.84$ $= 36.8$	
	(iii)	Air contains one-fifth oxygen by volume. What volume of air is required for the complete combustion of the 6 dm ³ of octane in the car engine? No of moles of oxygen required for combustion = $36.84 \times 25/2$ $= 460.5$ Volume of oxygen required = 460.5×24 $= 11052 \text{ dm}^3$ $= 11050 \text{ dm}^3$ Volume of air = 11050×5 $= 55250 \text{ dm}^3$ $= 55300 \text{ dm}^3$	
			[4]
	(b)	The fuel value (in kJ g ⁻¹) of a substance is the heat energy released when 1 g of the substance is combusted. Suggest, with relevant calculations, whether octane or ethanol, is a more economical fuel for automobiles.	
		Fuel value for octane = $5461/114 = 47.9 \text{ kJ g}^{-1}$ Fuel value for ethanol = $1366/46 = 29.7 \text{ kJ g}^{-1}$	
		Octane	
			[2]
	(c)	In the analysis of the exhaust gases, the percentages of gases do not add to 100%. What is the chief component of exhaust gases? Nitrogen gas.	
			[1]
	(d) (i)	Write equations to show how a catalytic converter converts the following gases to harmless products. • CO • NO	
		$\text{CO} + \frac{1}{2} \text{O}_2 \rightarrow \text{CO}_2$ $\text{NO} \rightarrow \frac{1}{2} \text{N}_2 + \frac{1}{2} \text{O}_2$	
	(ii)	State the type of reaction undergone for each of the equations you have written above. Reaction of CO: _____ Reaction of NO : _____ Reaction of CO is <u>oxidation</u>	

Reaction of NO is reduction

[3]

- (e) In a catalytic converter, there are two types of catalysts at work. Both types consist of a ceramic structure coated with a metal catalyst, usually platinum, rhodium and/or palladium.

The element platinum has a number of naturally occurring isotopes which are listed below.

Isotope	Relative isotopic mass	Natural abundance (%)
190	189.85	0.01
192	191.96	0.79
194	193.96	32.9
195	194.94	33.8
196	195.96	25.3
198	197.98	7.20

Use the data to calculate the relative atomic mass of platinum to **two** decimal places.

Relative atomic mass of Pt

$$= (189.85 \times 0.01 + 191.96 \times 0.79 + 193.96 \times 32.9 + 194.94 \times 33.8 + 195.96 \times 25.3 + 197.98 \times 7.20) / 100$$

$$= 195.07$$

[1]

[Total:11]

Section B (40 marks)

Answer **two** questions from this section on separate answer paper.

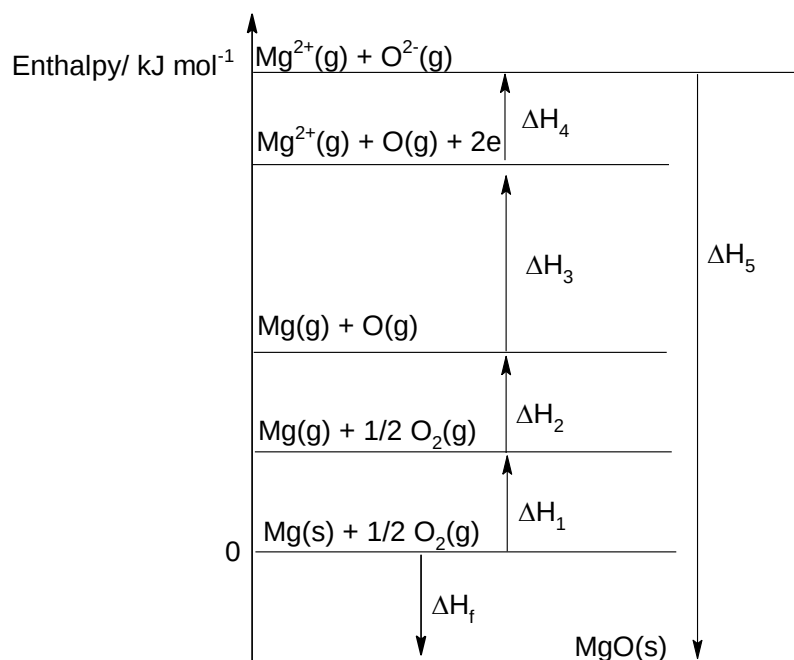
4	(a)	Describe what you would see when magnesium, phosphorus and sulfur are separately burned in air or oxygen. Write equations for the reactions that occur.	[5]
		Mg burns in oxygen with a bright/ <u>dazzling white flame</u> with a <u>white residue</u> produced. $\text{Mg (s)} + \frac{1}{2} \text{O}_2 \text{ (g)} \rightarrow \text{MgO (s)}$ P burns in oxygen with a <u>white flame</u> with a white residue produced. $4 \text{P (s)} + 3 \text{O}_2 \text{ (g)} \rightarrow \text{P}_4\text{O}_6 \text{ (s)}$ Or: $\text{P}_4 \text{ (s)} + 5 \text{O}_2 \text{ (g)} \rightarrow \text{P}_4\text{O}_{10} \text{ (s)}$ S burns in oxygen with a <u>blue flame</u>. $\text{S (s)} + \text{O}_2 \text{ (g)} \rightarrow \text{SO}_2 \text{ (g)}$	
	(b)	The oxides of phosphorus and sulfur resulting from the reactions in (a) react with water. Write equations for these two reactions and describe the effect of the resulting solutions on Universal Indicator solution.	[3]
		$\text{P}_4\text{O}_6 + 6 \text{H}_2\text{O} \rightarrow 4 \text{H}_3\text{PO}_3$ Or $\text{P}_4\text{O}_{10} + 6 \text{H}_2\text{O} \rightarrow 4 \text{H}_3\text{PO}_4$ Colour of universal indicator solution: <u>red / orange</u> $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$ Colour of universal indicator solution: <u>red</u>	
	(c)	Magnesium oxide has many applications in life. In the field of medicine, magnesium oxide is used for relief of heartburn and sore stomach. Gastric juice is an acidic digestive fluid secreted by glands in the mucous membrane that lines the stomach. It helps in the digestion of food in humans. However, over secretion of gastric juice would lead to gastric pain in humans. The gastric juice contains among other substances, 0.50% by mass of hydrochloric acid. An average adult produces about 36.5g of gastric juice when pain is experienced. An antacid is a substance which relieves pain by removing hydrochloric acid present in gastric juice. A typical 325 mg tablet of antacid usually contains 20 % by mass of magnesium oxide.	
	(i)	Write a balanced equation for the reaction between magnesium oxide and hydrochloric acid.	
		$\text{MgO(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{O(l)}$	
	(ii)	Calculate the number of antacid tablets an adult has to consume to alleviate the pain when 36.5g of gastric juice is produced in the stomach.	[3]
		Mass of HCl = $(36.5 \times 0.5/100) = 0.1825\text{g}$ Amount of HCl = $5.00 \times 10^{-3} \text{ mol}$	

Amt of MgO needed = $5 \times 10^{-3} / 2 = 2.50 \times 10^{-3}$ mol

Amt of MgO in one tablet = $(\frac{20}{100} \times 325 \times 10^{-3}) / 40.3 = 1.613 \times 10^{-3}$ mol

Number of tablets = $2.50 \times 10^{-3} / (1.613 \times 10^{-3}) = 1.55 = 2$

- (d) The formation of magnesium oxide from its elements may be represented by a Born-Haber energy cycle as described below.



- (i) Define the term *standard enthalpy change of formation* using magnesium oxide as an example.

Standard enthalpy change of formation is the heat change when one mole of MgO is formed from Mg and O₂ in their standard states at 298K and 1 atm.

- (ii) Name the enthalpy changes of ΔH_3 and ΔH_5 .

ΔH_3 is the sum of first and second ionisation energy of Mg.

ΔH_5 is the lattice energy of MgO

- (iii) Use values from the *Data Booklet* together with the values given below to calculate the enthalpy change of formation, ΔH_f , of magnesium oxide.

ΔH_1	+150 kJ mol ⁻¹
ΔH_5	-3889 kJ mol ⁻¹
$\Delta H_2 + \Delta H_4$	+950 kJ mol ⁻¹

$\Delta H_3 = 736 + 1450$
 $= + 2186 \text{ kJ mol}^{-1}$

By Hess' Law,

$\Delta H_f = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4 + \Delta H_5$

$$= 150 + 950 + 2186 + (-3889)$$

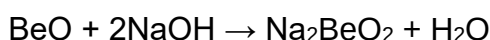
$$= -603 \text{ kJ mol}^{-1}$$

- (iv) The value of ΔH_5 for barium oxide is $-3152 \text{ kJ mol}^{-1}$. Account for the difference in the values of ΔH_5 for magnesium oxide and barium oxide. [7]

The lattice energy of MgO is more exothermic than BaO. This is because MgO has a stronger ionic bond between Mg^{2+} and O^{2-} compared to that between Ba^{2+} and O^{2-} .

Strength of ionic bonds $\propto |\text{LE}| \propto \left| \frac{q_+ \times q_-}{r_+ + r_-} \right|$
Ionic radius of $\text{Mg}^{2+} < \text{ionic radius of Ba}^{2+}$ while the anionic radius and the charges are the same. Hence, more energy is released when 1 mole of MgO is formed from the gaseous ions.

- (e) Beryllium is in the same group as magnesium. In the following reaction, discuss whether its behaviour is what you would expect from the position of the element in the Periodic Table. [2]



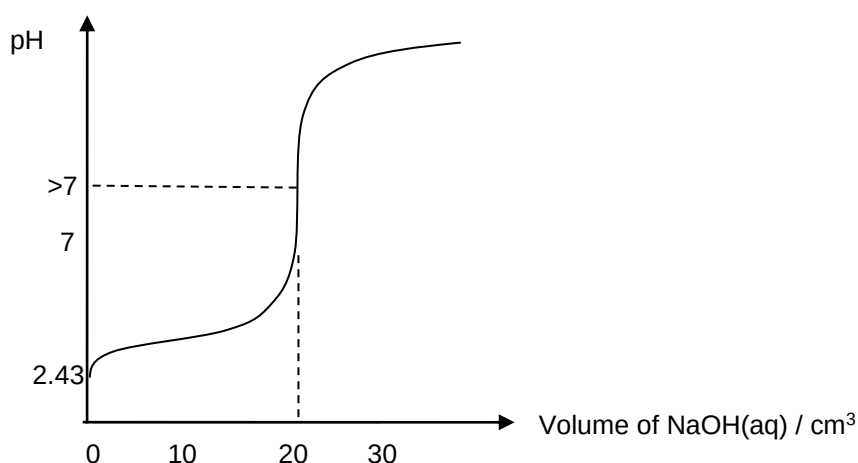
This acidic behaviour of BeO is not expected.

Be (Group II element) is metallic in character and is expected to form basic oxide. However, due to the high charge density of Be^{2+} ion, BeO shows high degree of covalency and is amphoteric (like Al_2O_3).

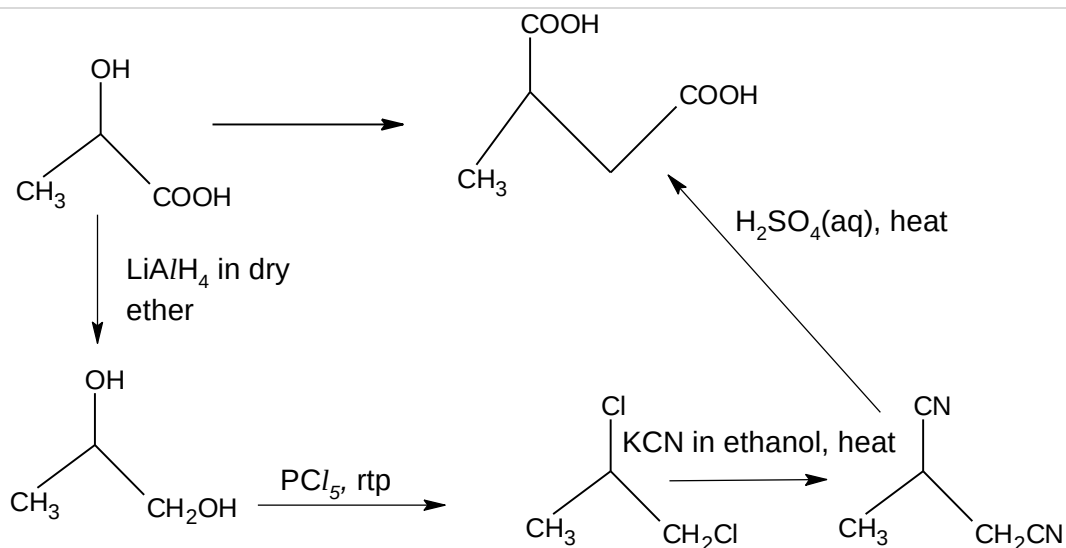
[Total:20]

- 5 (a) Lactic acid, $\text{C}_3\text{H}_6\text{O}_3$, which is also known as a milk acid is found primarily in milk products such as yogurt and cottage cheese.

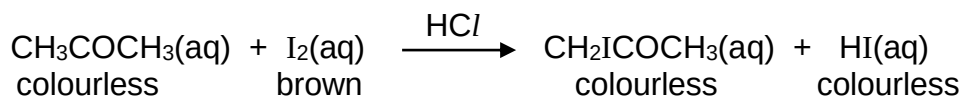
A 25.0 cm^3 sample of $0.100 \text{ mol dm}^{-3}$ lactic acid was titrated with a standard solution of $0.125 \text{ mol dm}^{-3}$ of aqueous sodium hydroxide. By using a pH meter, the following graph was obtained.



		(i)	By means of calculation, determine the number of acidic hydrogen atoms in one molecule of lactic acid.
			$n(\text{lactic acid}) = 0.00250 \text{ mol}$ $n(\text{NaOH}) = 0.00250 \text{ mol}$ $1 \text{ mol lactic acid} \equiv 1 \text{ mol NaOH}$ Lactic acid is a monocarboxylic acid and contains <u>one acidic hydrogen per molecule</u> .
		(ii)	Is lactic acid a weak or strong acid? Explain your answer.
			$\text{pH} = -\lg [\text{H}^+] = 2.43$ $[\text{H}^+] = 3.715 \times 10^{-3} \text{ mol dm}^{-3}$ $[\text{H}^+] \ll [\text{lactic acid}] \text{ or } 0.100 \text{ mol dm}^{-3}$, Lactic acid ionises partially and is a weak acid
		(iii)	Write an expression for the acid dissociation constant K_a for lactic acid. Hence, determine its K_a value. [5]
			$K_a = \frac{[\text{C}_3\text{H}_5\text{O}_3^-][\text{H}^+]}{[\text{C}_3\text{H}_6\text{O}_3]}$ $= \frac{0.003715^2}{0.10 - 0.003715}$ $= 1.43 \times 10^{-4} \text{ mol dm}^{-3}$
		(b)	Suggest how the following transformation can be achieved. Indicate reagents and conditions you would use for each step. Give structural formulae of the intermediate compounds formed. [5]



- (c) The kinetics of the acid catalysed reaction of propanone with iodine can be investigated experimentally by varying the concentrations of the three reactants involved and determining the time for the colour of the iodine to disappear.



In this method, the rate of the reaction is measured in terms of the rate at which the iodine concentration changes as shown below.

$$\text{Rate of reaction} \propto \frac{\text{volume of iodine used}}{\text{time taken for the colour of iodine to disappear}}$$

The following results were obtained from the experiment.

Experiment	Volume of solution / cm ³				Time taken for the brown iodine to disappear / s
	HCl	CH ₃ COCH ₃	H ₂ O	I ₂	
1	10	8	10	2	34
2	5	8	15	2	68
3	10	4	14	2	68
4	10	8	11	1	17

- (i) Calculate the rate for each of these four experiments.

Experiment	Volume of solution / cm ³				Rate
	HCl	CH ₃ COCH ₃	H ₂ O	I ₂	
1	10	8	10	2	0.0588
2	5	8	15	2	0.0294
3	10	4	14	2	0.0294
4	10	8	11	1	0.0588

(ii) Use the results in (c)(i) to deduce the order of reaction with respect to each of the three reactants in the reaction.

When concentration / vol of CH_3COCH_3 increased 2 times from experiment 3 to 1, keeping the concentration of I_2 and HCl constant, initial rate increased 2 times from $2/68$ to $2/34$, hence it is first order with respect to CH_3COCH_3 .

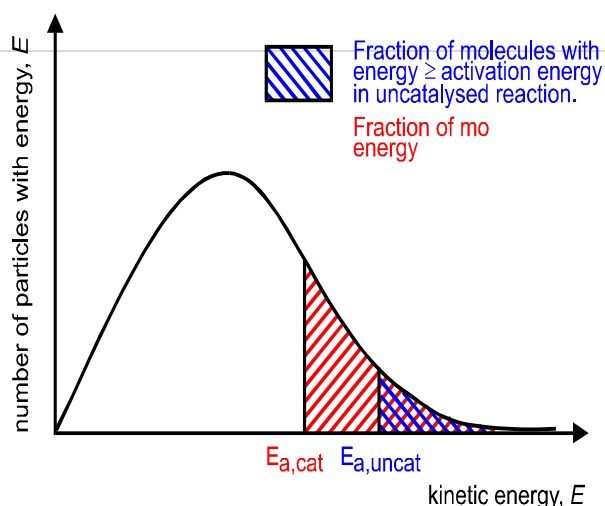
When concentration / vol of I_2 increased 2 times from experiment 4 to 1, keeping the concentration of CH_3COCH_3 and HCl constant, initial rate is unchanged at $1/17$, hence it is zero order with respect to I_2 .

When concentration / vol of HCl increased 2 times from experiment 2 to 1, keeping the concentration of CH_3COCH_3 and I_2 constant, initial rate increased by two times, hence it is first order with respect to HCl .

(ii) Hence, write an overall rate equation, stating the units of the rate constant.

rate = $k[\text{CH}_3\text{COCH}_3][\text{HCl}]$
Units: $\text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$

(iii) By using the concept of activation energy, together with an appropriate sketch of the Boltzmann distribution, explain why the addition of catalyst speeds up a chemical reaction. [10]



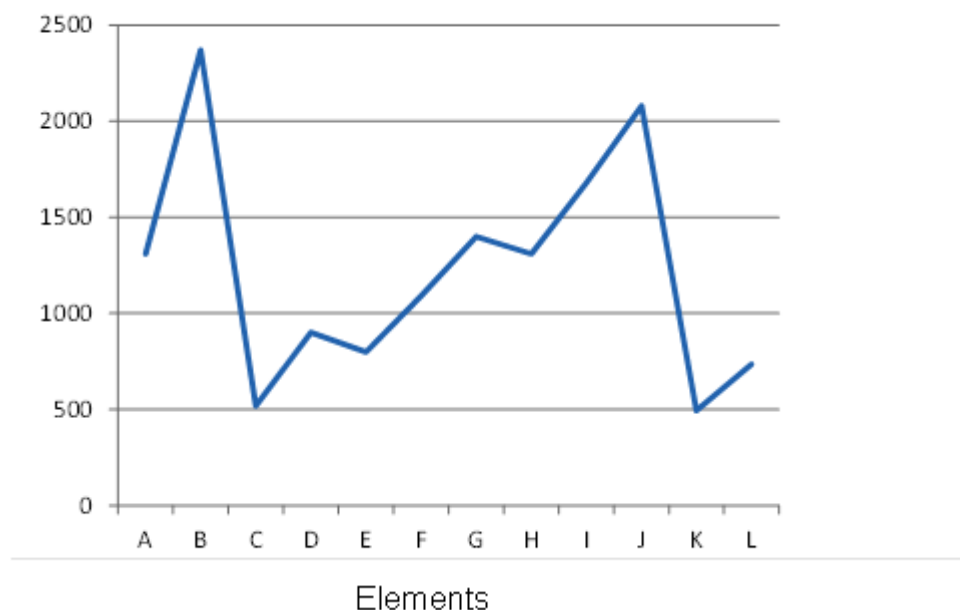
when a catalyst is used in a reaction, it:

- increases the rate of the reaction by providing an alternative reaction pathway with lower activation energy,
- increases the number of reacting particles with energy $\geq E_a$,
- (there is **no** increase in frequency of collisions between reacting particles!)
- increases the number of effective collisions per unit time

[Total:20]

- 6 (a) The following graph shows the trend of first ionisation energies of consecutive elements in the periodic table.

Ionisation Energies / kJ mol^{-1}



- (i) State and explain the general trend of first ionisation energies from element **C** to **J**.

Element C to J belong to elements in the same period. Ionisation energy generally increases across the period as the nuclear charge increases, while shielding effect remains constant. This would result in an increase in the effective nuclear charge, hence requiring more energy to remove an electron going across the period.

- (ii) Predict and explain whether the second ionisation energy of element **I** is lower or higher than that of element **H**.

The 2nd I.E. of element I is lower than element H.

I is in group VII.

I^+ : $ns^2 np^4$

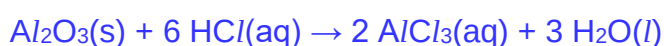
H^+ : $ns^2 np^3$

The electrons to be removed in both H and I are located in the p orbitals. However, the electron removed from I is from an orbital which contains 2 electrons, which experiences inter-electron repulsion. Less energy is required to remove the outermost electron of I.

- (iii) Element **H** exist as a gas at room temperature and pressure. It reacts readily with aluminium to form compound, **Z**.

Suggest an identity for compound **Z** and write equations for its reaction with HCl and NaOH .

Z is Al_2O_3 .





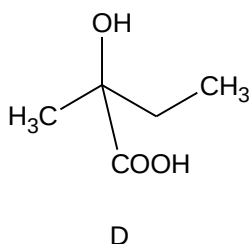
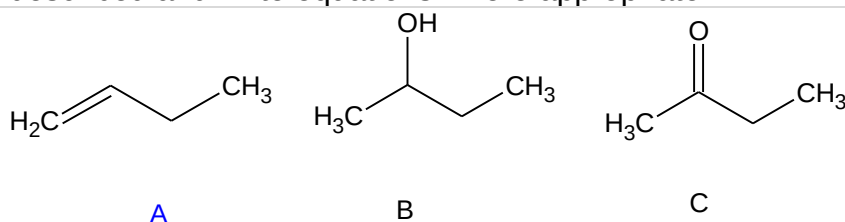
(iv) State the pH when compound **Z** is mixed with water. Explain your answer.

[10]

pH= 7 as aluminium oxide is insoluble in water.

- (b) An organic compound **A** gives off carbon dioxide gas upon reaction with hot acidified KMnO_4 . When reacted with steam, compound **A** forms compound **B**. Compound **B** reacts with sodium metal to evolve a gas that extinguishes a lighted splint with a pop sound. Compound **B** undergoes oxidation to form compound **C**, $\text{C}_4\text{H}_8\text{O}$. When compound **C** is warmed with alkaline aqueous iodine, a yellow precipitate is formed. Upon reaction with HCN in trace amount of NaCN at 10°C , followed by heating with $\text{H}_2\text{SO}_4(\text{aq})$, compound **C** forms compound **D**, $\text{C}_5\text{H}_{10}\text{O}_3$.

Deduce the structures of compounds **A** to **D**. Explain the chemistry of the reactions described and write equations where appropriate. [10]



An organic compound **A** gives off carbon dioxide gas upon reaction with hot acidified KMnO_4 .

A contains alkene functional group.
A undergoes oxidation with hot KMnO_4 .
A has terminal $-\text{C}=\text{C}-$.

When reacted with steam, compound **A** forms compound **B**.

A undergoes addition reaction with steam to form **B**.

Compound **B** can react with sodium metal to evolve a gas that extinguishes a lighted splint with a pop sound.

B undergoes reduction when reacting with sodium, **B** is an alcohol.
Hydrogen gas liberated.

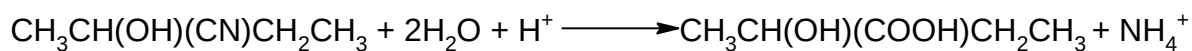
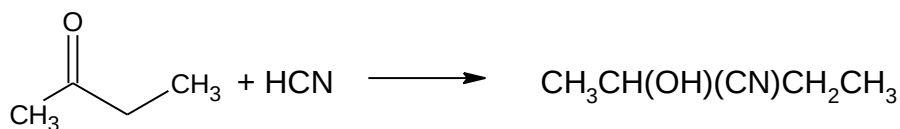
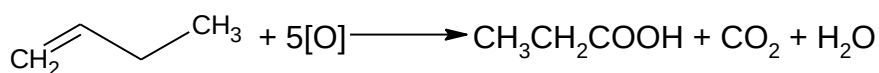
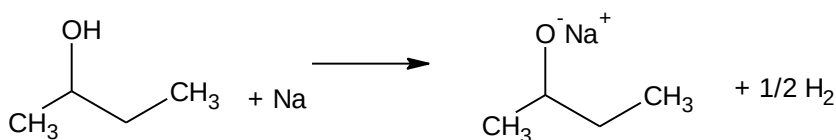
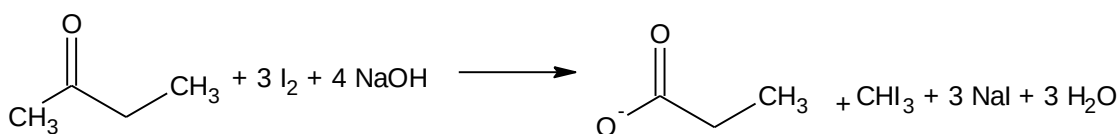
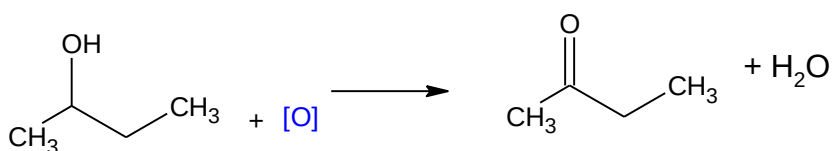
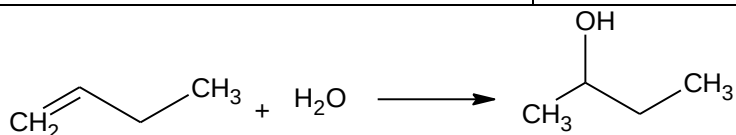
Compound **B** undergoes oxidation to form compound **C**, $\text{C}_4\text{H}_8\text{O}$

B is a 2° alcohol.
C is a ketone.

When compound **C** is warmed with alkaline aqueous iodine, a yellow ppt is formed.

C undergoes oxidation with alkaline aqueous iodine.

	$\begin{array}{c} \text{CH}_3 \\ \\ \text{C}=\text{O} \\ \\ \text{R} \end{array}$ It contains R Yellow ppt is CHI_3.
Upon reaction with HCN in trace amount of NaCN at 10°C , followed by heating with $\text{H}_2\text{SO}_4(\text{aq})$, compound C forms compound D, $\text{C}_5\text{H}_{10}\text{O}_3$.	C undergoes <u>addition</u> with HCN and <u>acidic hydrolysis</u> to form compound D. D contains <u>carboxylic acid</u> and <u>alcohol</u> group.



[Total: 20]

End of Paper